LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of corrosion on steel products, aluminum and magnesium alloys and copper alloys.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of corrosion on steel products, aluminum and magnesium alloys and copper alloys.
5. Specific Objectives	6.1.1.1 Trainee will be able to describe types of corrosion.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of corrosion definitions on steel, aluminum, magnesium, and copper alloys; Instructor supervision required for material handling safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the general definition of corrosion?
(b) Which metals are most susceptible to corrosion in aircraft?
(c) Why is understanding corrosion important for aircrew?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General definition of corrosion • Chemical process of oxidation • Environmental factors causing corrosion • Impact on aircraft structural integrity • Basic prevention concepts	L & Ds	PPT, IB	15 min
(b)	• Corrosion mechanisms in steel products • Corrosion mechanisms in aluminum alloys • Corrosion mechanisms in magnesium alloys • Corrosion mechanisms in copper alloys • Differences in alloy susceptibility	L & Ds	PPT, PDF	20 min
(c)	• Classification of corrosion types • Galvanic corrosion basics • Electrochemical cell formation • Role of electrolytes in corrosion • Initial identification of corrosion signs	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Define corrosion specifically for magnesium alloys.
(b) Explain how an electrolyte contributes to the corrosion process.
(c) Differentiate between corrosion types in steel versus aluminum.

12. Home Assignment and Follow Up	Prepare a brief note on the specific corrosion risks associated with copper alloys in aircraft systems, referencing the definitions learned in this lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various types of corrosion and their identification.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min